

MEng Project Worklog

Initial Software Reference Experiment for Neural Image Reconstruction

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Abstract

This worklog documents the first end-to-end software experiment for a proposed quantised FPGA accelerator for real-time neural image reconstruction. The objective was not to produce a final super-resolution model, but to establish a reproducible reference pipeline before hardware-oriented optimisation. The session verified CUDA execution on an NVIDIA GeForce RTX 3080, prepared isolated training and validation image pairs, implemented and verified a compact three-layer convolutional neural network (CNN), demonstrated learning on a single image pair, trained the network on a small dataset, and compared its held-out reconstruction quality against bilinear and bicubic interpolation. The pipeline completed successfully, but the initial CNN remained substantially below the interpolation baselines. This result motivates a controlled residual-learning experiment as the next step rather than premature quantisation or FPGA implementation.

1 Objective and Session Boundary

The software reference stage addresses a simple $2\times$ image-reconstruction task. A grayscale low-resolution image of size 128×128 is reconstructed to a 256×256 target. Three reconstruction methods are evaluated on identical held-out inputs:

1. bilinear interpolation;
2. bicubic interpolation; and
3. a compact CNN operating after bilinear $2\times$ upsampling.

The session boundary was deliberately restricted to the software reference implementation. FPGA integration, hardware purchases, quantisation, and DLSS-related work were deferred until the reconstruction and evaluation pipeline had been demonstrated to function correctly.

2 Environment and Reproducibility

Table 1: Verified execution environment.

Component	Recorded configuration
Python	3.13.1
PyTorch	2.14.0+cu130
CUDA runtime used by PyTorch	13.0
GPU	NVIDIA GeForce RTX 3080
GPU execution	PASS

The CUDA installation was verified with an explicit matrix multiplication on device `cuda:0`, rather than relying only on device discovery. The model verification and training runs were subsequently executed on the RTX 3080.

For the short training experiment, the configured random seed was 42, batch size was 8, learning rate was 10^{-3} , optimizer was Adam, and the run length was 30 epochs. Checkpoints and CSV histories were preserved in the experiment results.

3 Dataset and Preprocessing

For the bring-up experiment, a small subset of DIV2K was used. Twelve source images (0001–0012) formed the training source set and four separate validation-source images (0801–0804) formed the held-out validation set. The images were downloaded as JPEG copies from the Benjamin-Wegener/DIV2K GitHub mirror. The split was performed at the source-image level before patch extraction, preventing patches from the same photograph from appearing in both training and validation data.

Table 2: Prepared reconstruction dataset.

	Training	Validation
Source images	12	4
Generated pairs	60	4
HR patch size	256×256	256×256
LR input size	128×128	128×128
Colour representation	grayscale	grayscale
Pixel representation	$[0, 1]$	$[0, 1]$

Five deterministic 256×256 patches were extracted per training source image (four corners and centre), producing 60 training pairs. One deterministic centre crop was used from each validation source, producing four held-out pairs. Each HR patch was downsampled to 128×128 using bicubic interpolation, providing a fixed $2\times$ degradation process.

Important baseline note. An earlier one-image smoke test used the original baseline script and produced the interpolation results shown in Table 3. That first script used a different preliminary downsampling choice and is therefore retained only as a sanity check. The later held-out comparison is the scientifically relevant comparison because the CNN, bilinear, and bicubic methods are all evaluated from the same fixed validation inputs.

4 Evaluation Metrics

Mean squared error (MSE) between a reference image I and reconstruction $\hat{I}$ is

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N \left(I_i - \hat{I}_i \right)^2. \quad (1)$$

With pixels normalised to $[0, 1]$, peak signal-to-noise ratio (PSNR) is

$$\text{PSNR} = 10 \log_{10} \left(\frac{1}{\text{MSE}} \right) \text{ dB}. \quad (2)$$

Lower MSE and higher PSNR indicate better pixel-wise reconstruction fidelity.

Table 3: Initial one-image interpolation smoke test.

Method	MSE	PSNR (dB)
Bilinear	0.00045050	33.463
Bicubic	0.00033016	34.813

5 Compact CNN Architecture

The first neural model was deliberately small to provide an FPGA-friendly reference point and to minimise debugging complexity. The processing chain was

$$128 \times 128 \xrightarrow{\text{bilinear } 2\times} 256 \times 256 \xrightarrow{3\times 3 \text{ conv}} 8 \xrightarrow{\text{ReLU}} 8 \xrightarrow{3\times 3 \text{ conv}} 8 \xrightarrow{\text{ReLU}} 8 \xrightarrow{3\times 3 \text{ conv}} 1.$$

The convolutional channel sequence was $1 \rightarrow 8 \rightarrow 8 \rightarrow 1$, with 3×3 kernels and padding chosen to preserve the 256×256 spatial dimensions.

The trainable parameter count was

$$N_\theta = (1 \cdot 8 \cdot 3 \cdot 3 + 8) + (8 \cdot 8 \cdot 3 \cdot 3 + 8) + (8 \cdot 1 \cdot 3 \cdot 3 + 1) \quad (3)$$

$$= 737. \quad (4)$$

6 Forward and Backward Verification

Before broader training, one training pair was passed through the network. The following checks all passed:

- LR input shape: $(1, 1, 128, 128)$;
- HR target shape: $(1, 1, 256, 256)$;
- CNN output shape: $(1, 1, 256, 256)$;
- MSE loss calculation completed;
- backpropagation completed;
- convolutional weights changed after the optimizer step; and
- model output and loss remained finite.

The initial verification loss fell from 0.11031403 to 0.10197250 after one parameter update. This was treated only as a computational-graph verification, not as evidence of reconstruction quality.

7 Single-Pair Learning Check

The network was then deliberately trained repeatedly on one fixed LR/HR pair for 500 steps. This is an implementation test: a model that cannot substantially reduce error on one example should not be trusted in a larger training experiment.

Table 4: Single-pair learning result.

Metric	Initial	Final
MSE	0.20750067	0.00064742
PSNR (dB)	6.830	31.888

The MSE decreased by approximately $320.5\times$, while PSNR improved by 25.058 dB. The loss decreased smoothly throughout the 500-step run and no invalid numerical values were observed.

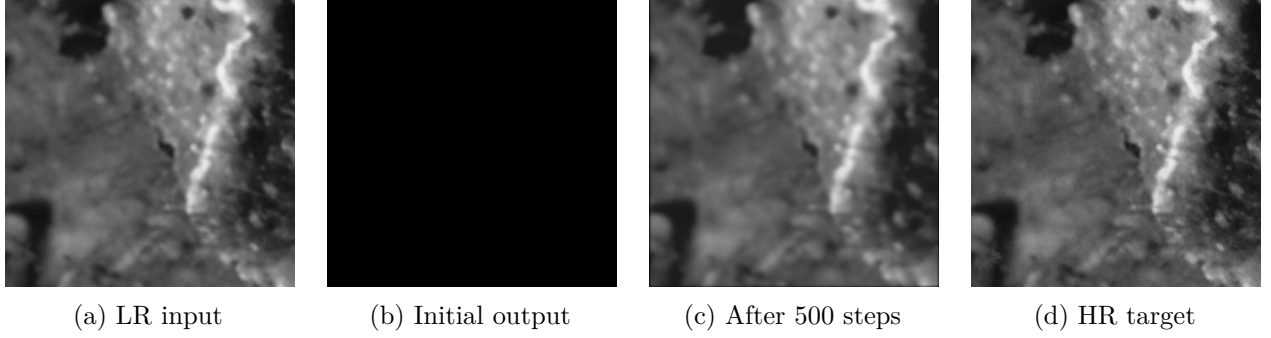


Figure 1: Single-pair implementation check. This result demonstrates learnability of the implementation, not generalisation.

8 Short Training Experiment

A fresh model was initialised before the 30-epoch training run. All 60 training pairs were used for gradient updates, while the four validation pairs remained excluded from optimisation.

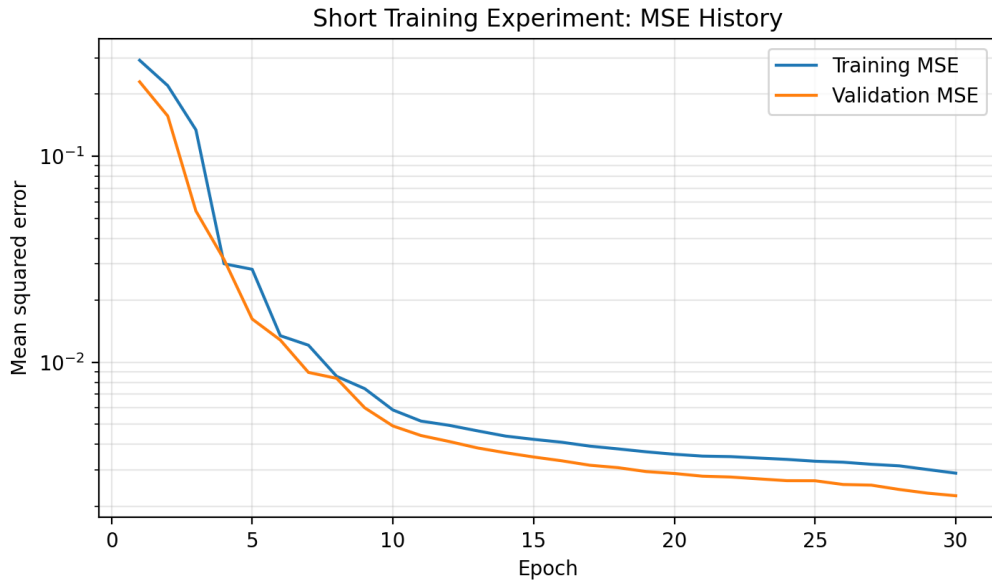


Figure 2: Training and validation MSE over 30 epochs. A logarithmic vertical axis is used to show the large early reduction in loss.

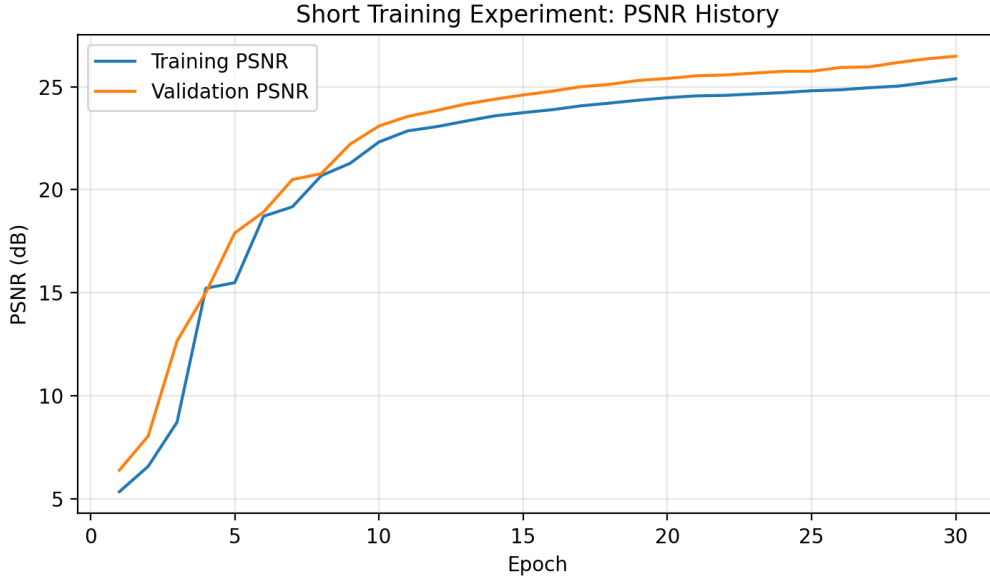


Figure 3: Training and validation PSNR over the short training run.

The best validation result occurred at epoch 30, with validation MSE 0.00224876 and PSNR 26.481 dB. The validation loss continued to improve at the end of the run, so the experiment did not show the conventional pattern of a training loss continuing to fall while validation loss degraded. However, the validation set contains only four samples, so this observation should not be interpreted as strong statistical evidence against overfitting.

9 Held-Out Comparison

All three reconstruction methods were evaluated on the same four validation LR inputs and the same HR targets. Table 5 reports the raw per-image metrics.

Table 5: Held-out validation metrics.

Image	CNN		Bilinear		Bicubic	
	MSE	PSNR	MSE	PSNR	MSE	PSNR
0801	0.00271219	25.667	0.00034383	34.637	0.00024150	36.171
0802	0.00151592	28.193	0.00023949	36.207	0.00016923	37.715
0803	0.00152638	28.163	0.00035044	34.554	0.00019694	37.057
0804	0.00324052	24.894	0.00141457	28.494	0.00112284	29.497

The original comparison script first averaged MSE across the four images and then converted that pooled MSE into a PSNR value. For transparency, Table 6 reports both that script-compatible aggregate and the arithmetic mean of the four per-image PSNR values. The ranking is unchanged under either convention.

Table 6: Aggregate held-out comparison.

Method	Mean MSE	PSNR from mean MSE (dB)	Mean per-image PSNR (dB)
CNN	0.00224875	26.481	26.729
Bilinear	0.00058708	32.313	33.473
Bicubic	0.00043263	33.639	35.110

Using the script-compatible aggregate, the CNN MSE was approximately $3.83\times$ the bilinear MSE and $5.20\times$ the bicubic MSE. The CNN trailed bilinear by 5.83 dB and bicubic by 7.16 dB. Bicubic reduced aggregate MSE by approximately 26.3% relative to bilinear and improved the corresponding PSNR by 1.33 dB.

9.1 Representative Validation Case: 0801

Image 0801 illustrates a case in which interpolation is already strong. Bicubic achieved 36.171 dB, bilinear 34.637 dB, and the CNN 25.667 dB.

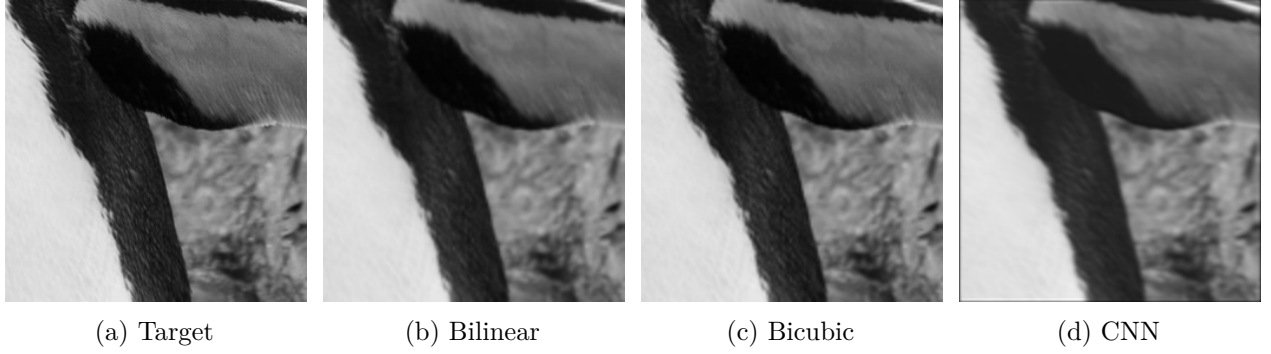


Figure 4: Held-out reconstruction comparison for validation image 0801.

9.2 Harder Validation Case: 0804

Image 0804 produced the lowest PSNR for all three methods and is therefore a useful stress case. The CNN achieved 24.894 dB, compared with 28.494 dB for bilinear and 29.497 dB for bicubic.

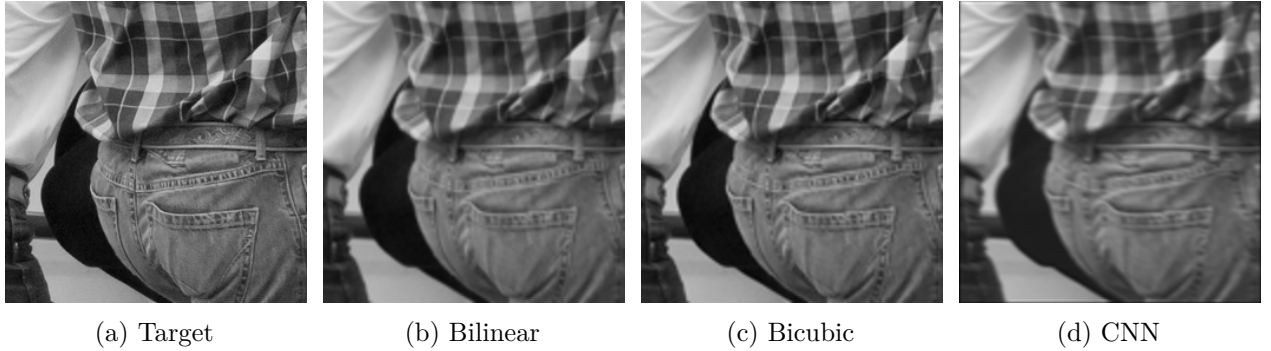


Figure 5: Held-out reconstruction comparison for validation image 0804.

10 Engineering Interpretation

The session established a functioning end-to-end reference pipeline. The CNN successfully learned a reconstruction mapping, the single-pair test showed strong optimisation behaviour, and held-out validation performance improved throughout the short training run. Therefore, the software implementation is sufficiently functional to support controlled architecture experiments.

The first CNN is nevertheless not yet a useful reconstruction model relative to conventional interpolation. This is not unexpected: the model contains only 737 parameters, was trained on 60 patches, and is asked to produce the entire reconstructed image after a bilinear upsampling stage. The held-out evidence shows that both interpolation baselines remain decisively superior, with bicubic best on every validation image.

Visual inspection is consistent with the numerical result. In image 0801, the CNN softens the high-contrast boundary and fine background texture more strongly than the interpolation outputs.

In the harder 0804 case, fabric checks, seams, pocket edges, and other local texture are visibly smoothed in the CNN reconstruction. No clear compensating detail recovery was observed that would justify the lower PSNR.

The appropriate response is not to move directly to quantisation. Quantising a model that is already substantially below the baseline would preserve an algorithmic weakness in a hardware-friendly form. The next experiment should first improve the software reference while changing as few variables as possible.

11 Limitations

- The training set is intentionally small: 12 source images and 60 extracted patches.
- The held-out validation set contains only four centre crops, so aggregate metrics have high sampling uncertainty.
- The bring-up dataset uses JPEG copies from a GitHub mirror rather than the original lossless DIV2K distribution; thesis-quality experiments should use a documented canonical dataset release.
- Only MSE and PSNR were evaluated. They quantify pixel fidelity but do not fully describe perceptual reconstruction quality.
- The compact CNN was chosen for implementation simplicity and FPGA relevance, not because it represents a competitive modern super-resolution architecture.
- The first one-image baseline smoke test used a preliminary degradation path and should not be combined numerically with the final held-out experiment.

12 Next Controlled Experiment: Residual Reconstruction

The next model should retain the same data split, degradation process, metrics, and evaluation code while changing only the reconstruction formulation. Instead of predicting the full HR image, the CNN should predict a residual correction to a bilinear baseline:

$$R = I_{\text{HR}} - I_{\text{bilinear}}, \quad (5)$$

$$\hat{I} = I_{\text{bilinear}} + R_{\theta}. \quad (6)$$

This formulation gives the small network a narrower task: learn the high-frequency correction that interpolation misses rather than regenerate low-frequency image content already available in the bilinear reconstruction. It also remains attractive for later FPGA mapping because it requires only convolution, ReLU, and element-wise addition around a deterministic upsampling path.

The residual model should be accepted for further work only after it is evaluated on the same four held-out inputs and compared directly with the existing results in Table 5. If reconstruction quality becomes competitive, the subsequent step is controlled quantisation and measurement of the accuracy-resource trade-off.

13 Definition of Done

Requirement	Status
Reproducible training/validation pairs	PASS
Working CNN forward and backward passes	PASS
Recorded GPU execution status	PASS
Single-pair learning check	PASS
Preliminary held-out validation comparison	PASS
Saved checkpoint, metrics, images, and worklog	PASS

Evidence Files

The LaTeX package accompanying this report includes the original `worklog.txt`, `baseline_metrics.txt`, `training_history.csv`, and `comparison_metrics.csv` used to produce the reported values. Representative reconstruction images and generated training plots are stored under `figures/`.